

Pantufo Data

DATA SCIENCE

[TECHNIQUES]

[GUIDE WORKFLOW]

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DATA SCIENCE WORKFLOW

Data Management and Preparation

Before analysis can begin, data must be collected, cleaned, and prepared.

- **Data Collection/Sourcing:** Gathering raw data from various sources, which may include databases, APIs, or data streams.
- **Data Cleaning/Wrangling/Preprocessing:** Handling inconsistent data, missing values imputation, removing outliers, and transforming data into a usable format.
- **Exploratory Data Analysis (EDA):** The initial process of analyzing data sets to summarize their main characteristics, often with visual methods.
- **Feature Selection/Dimensionality Reduction:** Techniques like Principal Component Analysis (PCA) used to reduce the number of variables in a dataset while retaining important information.

Analytical and Modeling Techniques

These are the core methods for extracting insights and building predictive systems. Statistical Techniques

- **Descriptive Statistics:** Methods for summarizing data, such as mean, median, mode, and standard deviation.
- **Inferential Statistics:** Techniques like hypothesis testing and confidence intervals to make predictions or draw conclusions about a population based on a sample.
- **Regression Analysis:** Modeling relationships between variables to make predictions (e.g., linear regression, logistic regression).

Machine Learning (ML)

ML algorithms are used to build models that learn from data and make predictions or decisions.

- **Supervised Learning:** Training algorithms on labeled data.
 - **Classification:** Categorizing data into predefined classes (e.g., spam filtering, image recognition) using methods like **Decision Trees**, **Naïve Bayes**, **Support Vector Machines (SVM)**, and **Neural Networks**.
 - **Regression:** Predicting continuous numerical values.
- **Unsupervised Learning:** Identifying patterns in unlabeled data.
- **Clustering:** Grouping similar data points together using methods like **K-means Clustering**, **Mean-shift**, and **DBSCAN**.

- **Semi-supervised Learning:** Combining labeled and unlabeled data.
- **Reinforcement Learning:** Training agents to make decisions in an environment through rewards and penalties.
- **Ensemble Learning:** Combining multiple models (e.g., **Random Forests**) to improve predictive accuracy.

Specialized Domain Techniques

- **Natural Language Processing (NLP) / Text Mining:** Enabling machines to understand and interpret human language (e.g., sentiment analysis, chatbots).
- **Time Series Analysis:** Analyzing time-ordered data points for forecasting (e.g., stock market analysis).
- **Anomaly Detection:** Identifying rare items or outliers in data, often used for fraud detection.
- **Recommender Systems:** Providing personalized recommendations based on user behavior (e.g., collaborative filtering).
- **Simulation:** Using techniques like Monte Carlo Simulation to model probability and risk.

Communication and Deployment

The final steps involve sharing findings and putting models into practice.

- **Data Visualization:** Representing data graphically through charts, graphs, and dashboards to communicate insights effectively.
- **Storytelling:** Explaining data-driven insights in business-relevant terms to non-technical audiences.
- **Model Deployment (MLOps):** Integrating machine learning models into production environments so they can make real-time predictions.

TOOLS

Data science tools cover programming (**Python, R, SQL**), data manipulation (**Pandas, Spark, Excel**), visualization (**Tableau, Power BI, Matplotlib**), machine learning (**Scikit-learn, TensorFlow, PyTorch**), big data (Hadoop, Spark), cloud (**AWS, Azure, GCP**), databases (**SQL, MongoDB**), and MLOps/DevOps (**Jupyter, Git, Docker**). These tools facilitate tasks from data cleaning and analysis to model building and deployment, with Python and its libraries (**NumPy, Pandas, Scikit-learn**) being central to the ecosystem.

Core Languages & Environments

- Python: The most popular language for data science, with extensive libraries.
- R: Strong for statistical computing and graphics.
- SQL: Essential for database querying and management.
- Jupyter Notebooks: Interactive environment for coding and visualization.

Data Wrangling & Analysis Libraries (Python)

- Pandas: For data manipulation and analysis (DataFrames).
- NumPy: For numerical operations, especially arrays.
- Polars: A fast alternative to Pandas for larger datasets (emerging).
- Dplyr (R): For data manipulation in the R ecosystem.

Machine Learning & AI Frameworks

- Scikit-learn: Classic ML algorithms in Python.
- TensorFlow & PyTorch: Deep learning frameworks.
- Keras: High-level API for deep learning (often used with TensorFlow).
- Hugging Face: For Natural Language Processing (NLP) models.

Data Visualization & BI Tools

- Tableau & Power BI: Industry-standard Business Intelligence (BI) tools for dashboards.
- Matplotlib & Seaborn (Python): For creating static and statistical plots.
- Plotly & D3.js: For interactive web-based visualizations.

Big Data & Cloud Platforms

- Apache Spark & Hadoop: For processing large datasets.
- Cloud Platforms (AWS, Azure, Google Cloud): Services like SageMaker, BigQuery, Redshift for scalable computing.

- Snowflake: Cloud data warehousing.

Databases & Storage

- SQL Databases (MySQL, PostgreSQL): Relational data storage.
- NoSQL (MongoDB, Cassandra): Flexible document/key-value storage.

Development & MLOps

- Git & GitHub/GitLab: Version control and collaboration.
- Docker & Kubernetes: Containerization and orchestration.
- MLflow & Airflow: For managing machine learning lifecycles and workflows.

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